

Advancing Controllable Summarization: A Strategic Prosdegic Proposal

From a Critical Deconstruction of *Salient Information Prompting* to a Novel Semantic and Multilingual Framework

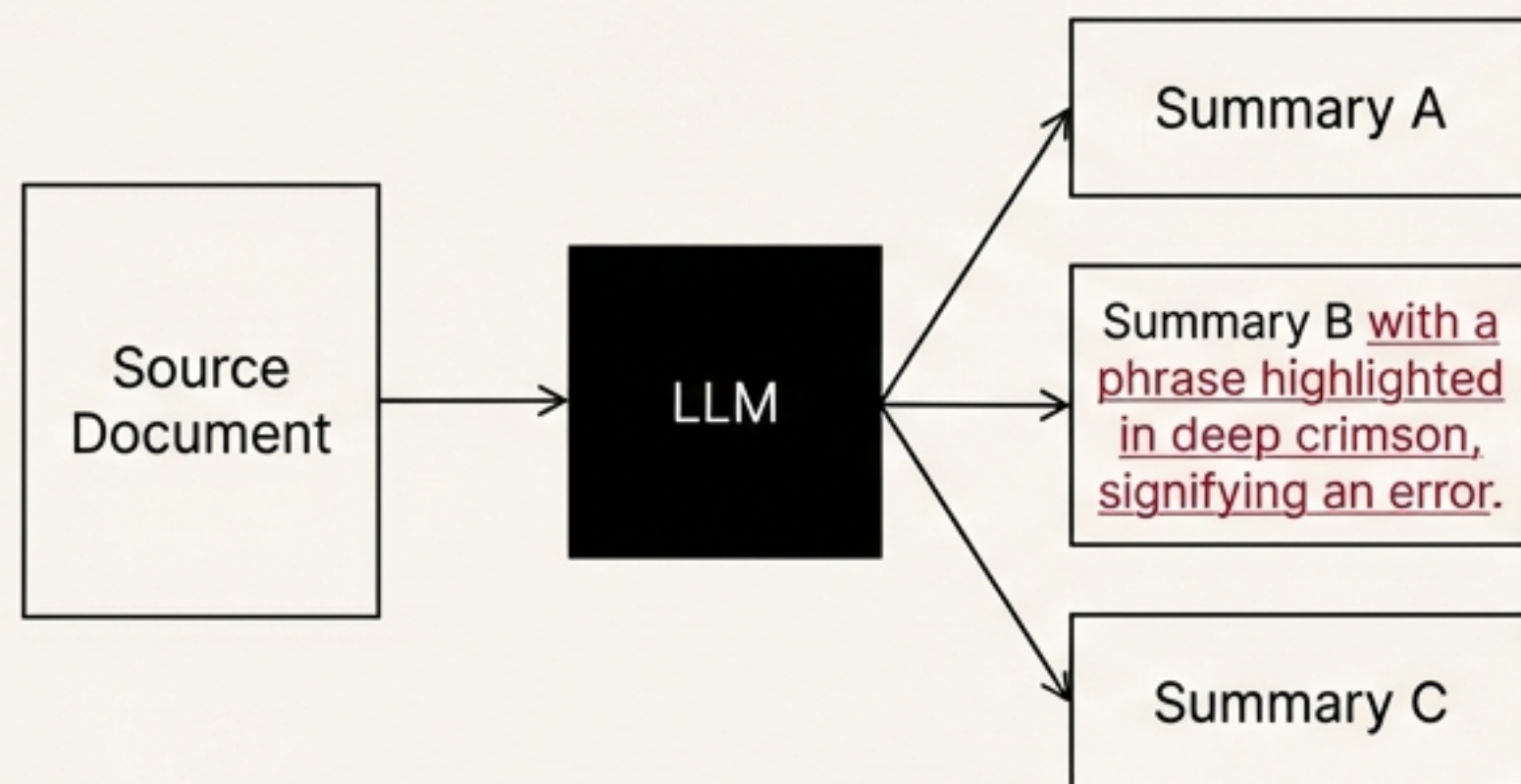
This project is designed to achieve the maximum score (10/10 + 5 extension points) and lay the groundwork for a potential scientific publication.

The Controllability Crisis in Modern Summarization

The Problem

The “Black Box” Dilemma

Large Language Models (GPT-4, Llama 3) generate fluent summaries but lack explicit control. Their internal determination of “salience” is opaque, leading to critical errors.



The Consequences

Two Critical Failure Modes



Omission of Critical Details:

“Models overlook statistically rare but contextually vital information.”

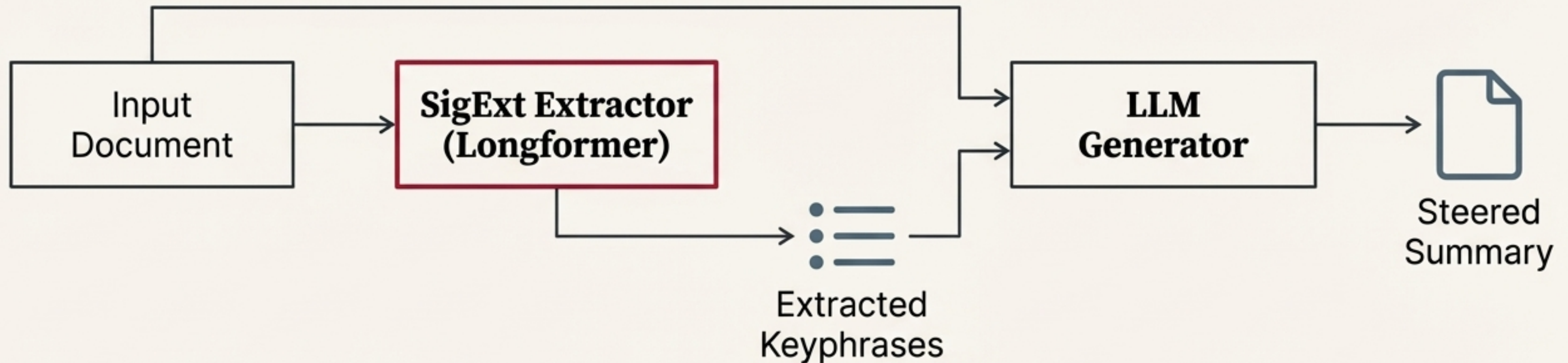


Factuality Hallucination:

“Models generate plausible but non-factual content, distorting relationships from the source text.”

The State-of-the-Art: Salient Information Prompting (SIP)

Xu et al. (2024) propose decoupling salience identification from generation. An **extractor** module finds key phrases *before* generation to steer the LLM.



Key Component: SigExt Extractor

The extractor, **SigExt**, is a supervised model built on **Longformer**, chosen for its ability to process long documents (up to 4,096 tokens) with linear $O(N)$ complexity.

The Achilles' Heel: A Foundation Built on Fuzzy Matching

How is the SigExt extractor trained? The paper's answer reveals a profound methodological weakness.

Heuristic Label Generation

To create training data, the authors label source sentences as "salient" if they have high character-level overlap with the reference summary.

$$Fuzz(p, q) = \frac{LongestCommonSubsequence(p, q)}{\max(|p|, |q|)}$$

If $Fuzz > 0.7$, label = 1.

Lexical Overlap $\neq$ Semantic Salience

This method fails when summaries use synonyms or paraphrasing. It trains the model to find string matches, not conceptual importance.

Killer Example

Source: "The company declared bankruptcy."

Summary: "The firm failed."



Result: Low Fuzzy Score. The model is incorrectly taught that the source sentence is *not* salient.

Navigating the Crossroads: Evaluating Our Initial Drafts

The Hallucination Paradox

The paper notes that for some LLMs (e.g., Mistral), adding keyphrases can **increase** hallucinations, as the model may use them out of context. This reinforces the need for a more robust approach.

Draft 1 (Decoder-Only / DecLLM)	Draft 2 (Encoder-Decoder / RedLLM)
Technology Llama 3, Mistral-7B (SOTA)	Technology BART, PEGASUS
Strengths Technologically current, RAG integration, Self-Refine concept.	Strengths Efficient, advanced metric ideas.
Fatal Flaw The proposed 'Precision-Recall Analysis' extension is not novel ; it merely replicates Figure 2 from the reference paper. This is a reproducibility task (2 points), not a 5-point extension.	Fatal Flaw Technologically dated, severe context limits (1024 tokens) unsuitable for long documents, methodologically inconsistent (unsupervised KeyBERT vs. supervised SigExt).

Draft 1 is the superior foundation, but it requires a genuinely innovative extension to secure the maximum score.

Our Resolution: The 'SM-SIP' Unified Project

Enhancing Prompt-Based Summarization via Semantically-Supervised
Keyphrase Extraction in Low-Resource Settings



Semantic Supervision

We replace the flawed Fuzzy Matching with a supervision signal based on true semantic similarity, teaching the extractor to understand concepts, not just words.

Addresses:
The "Achilles"
Heel" of the
original paper.

SM-SIP



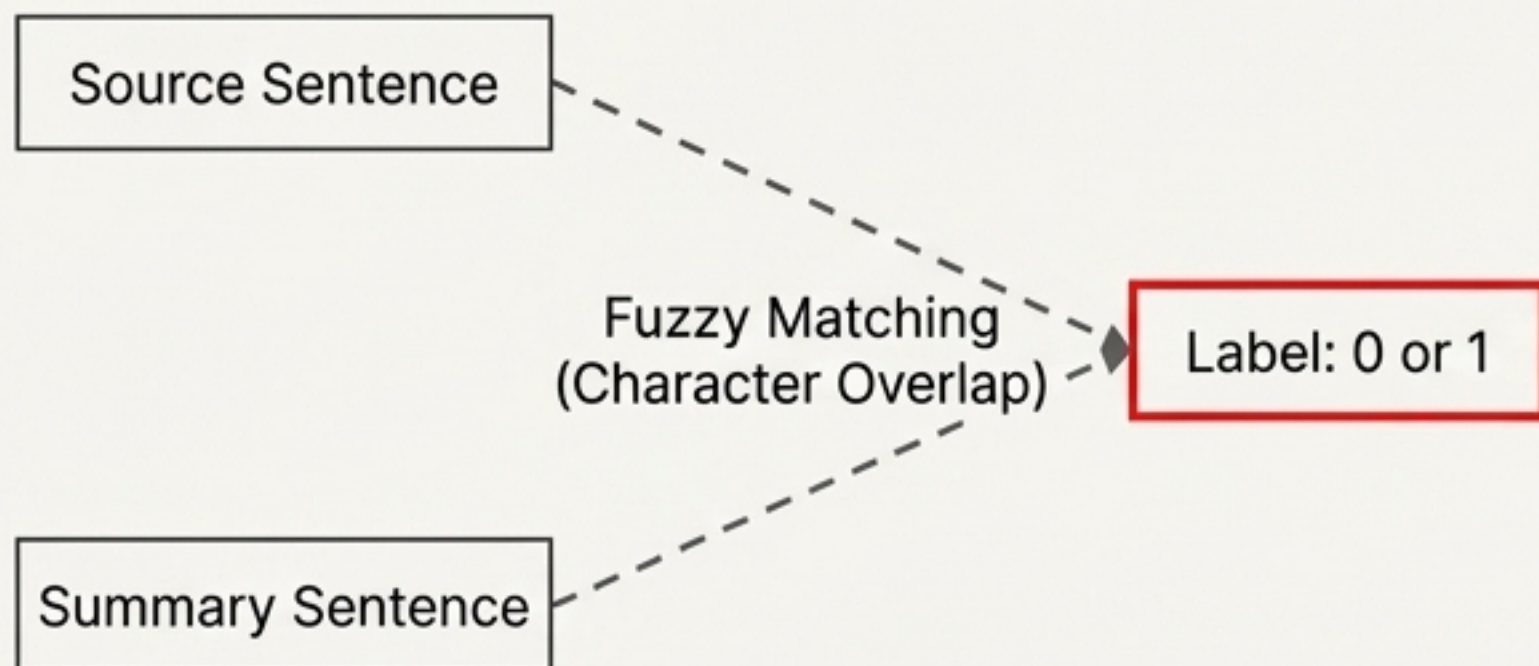
Multilingual Adaptation

We validate the SIP framework's robustness beyond English, adapting it to a morphologically rich, lower-resource language: Italian.

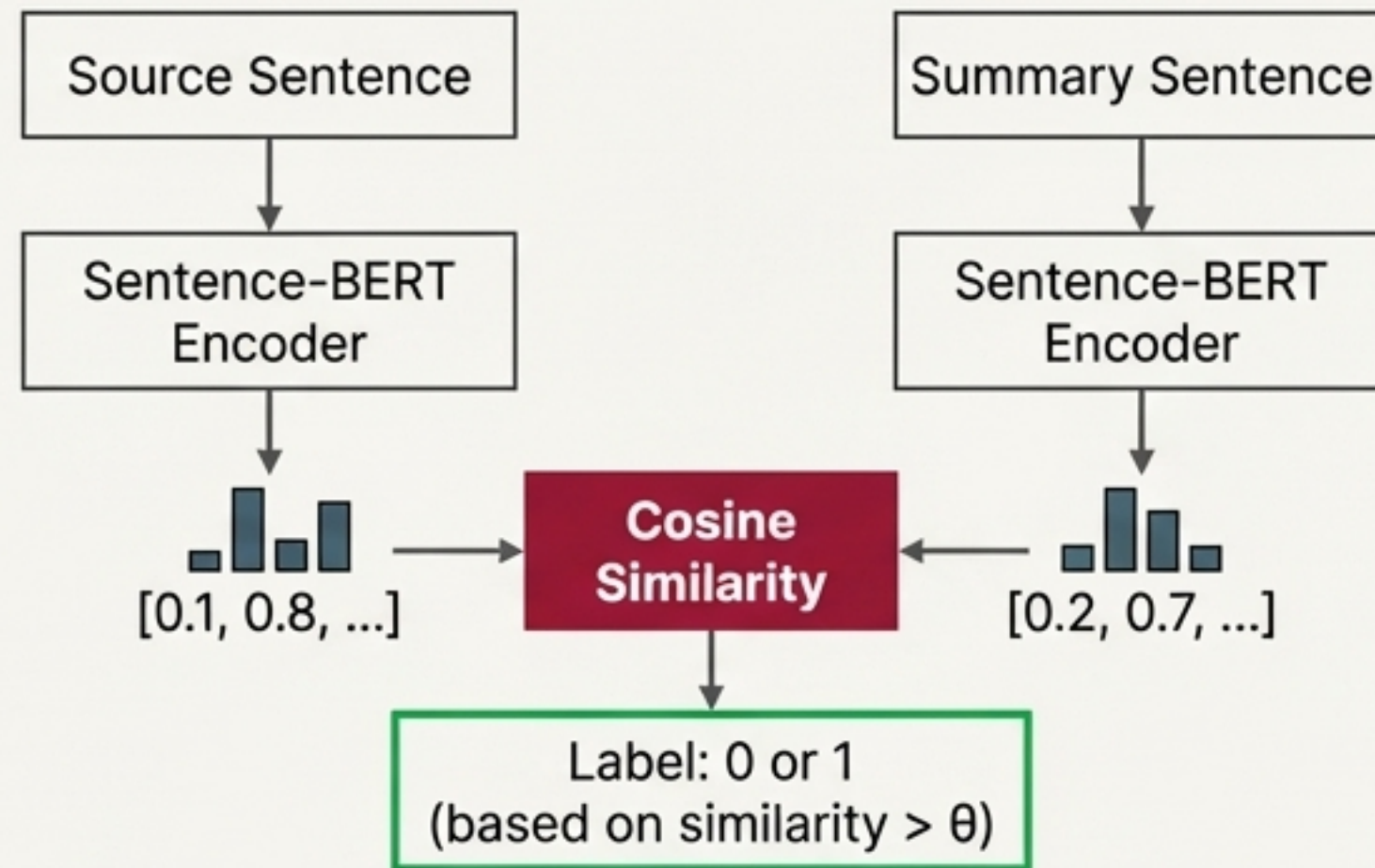
Addresses:
Course
requirements for
extensions and
scientific gap in
cross-lingual
validation.

Extension 1: From Lexical Guesswork to Semantic Supervision

Before (The Paper's Fuzzy Method)



After (Our Semantic Method)



Implementation Details & Scientific Value

- **Model:** `paraphrase-multilingual-mpnet-base-v2` (a Sentence-BERT model).
- **Hypothesis:** Training SigExt with semantic labels will produce a superior signal, leading to the extraction of more conceptually relevant keyphrases.

Scientific Value: This transforms the paradigm from lexical pattern matching to semantic concept identification—a novel contribution.

Extension 2: Proving Cross-Lingual Robustness in Italian

Demonstrate that the SIP method generalizes to morphologically richer languages and is not an English-specific artifact.

Key Challenge & Solution

- **Challenge:** No pre-trained Italian Longformer exists.
- **Solution:** We will use `markussagen/xlm-roberta-longformer-base-4096`, a multilingual model capable of handling long sequences, and fine-tune it.



Dataset Selection

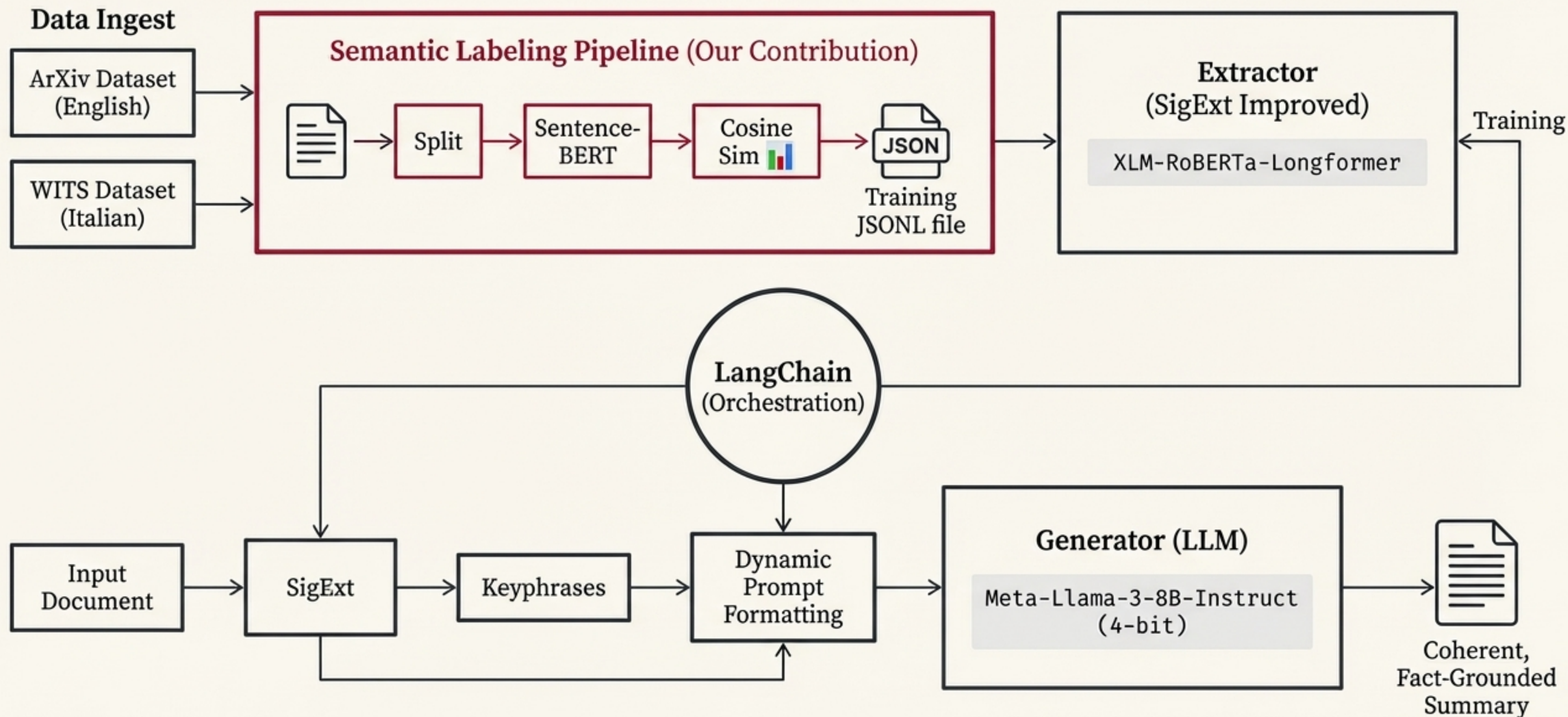
- Dataset: **WITS** (Wikipedia for Italian Text Summarization).
- Why WITS?: Its long, complex articles are the perfect testbed for a Longformer-based architecture, unlike shorter news datasets.



Scientific Value

1. Validates the SIP method in a lower-resource setting.
2. Tests the ability of Llama 3 to follow content constraints in Italian.
3. Creates a valuable new artifact: a SigExt model fine-tuned for Italian salient sentence extraction.

The SM-SIP Master Blueprint



A Rigorous Experimental Protocol

To empirically prove that our Semantic Supervision method is superior to both a zero-shot baseline and the original paper's Fuzzy Matching approach.

Baseline: Zero-Shot

Description: Llama 3 generates a summary of a WITS article with no keyphrase prompting.

Purpose: Establishes the performance floor.

Reproduction: SigExt-Fuzzy

Description: Llama 3 prompted with keyphrases from a SigExt model trained with the original paper's Fuzzy Matching labels (on Italian data).

Purpose: Replicates the paper's methodology for a fair comparison.

Our Extension: SigExt-Semantic

Description: Llama 3 prompted with keyphrases from our SigExt model trained with superior Semantic Similarity labels.

Purpose: Demonstrates the value of our core innovation.

Central Hypothesis: We expect SigExt-Semantic to significantly outperform SigExt-Fuzzy on metrics of semantic quality and steering effectiveness (BERTScore and KIR).

Evaluating Success Beyond Lexical Overlap

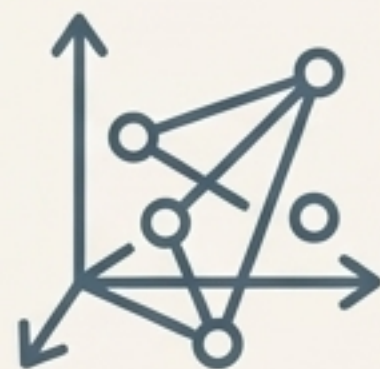
A top-tier project requires a holistic evaluation suite. We will measure performance across four distinct dimensions of quality.



Lexical Overlap
(The Standard)

ROUGE-1/2/L

Measures word/phrase overlap with the reference summary.



Semantic Similarity
(The Core of Our Claim)

BERTScore

Measures similarity of meaning using contextual embeddings. Crucial for validating our semantic extension.



Factual Faithfulness
(The Hallucination Check)

AlignScore / NLI

Whether the summary is factually entailed by the source document.



Steering Effectiveness
(The Control Check)

Keyphrase Inclusion Rate (KIR) - *Our custom metric*

The percentage of prompted keyphrases that appear in the final summary.

$$KIR = \frac{|\{k \in \text{Keyphrases} : k \in \text{Summary}\}|}{|\text{Keyphrases}|}$$

Projecting Our Success: Anticipated Results on the WITS Dataset

Method	ROUGE-1	BERTScore	KIR (Inclusion)	Notes
Llama-3 Zero-Shot	0.35	0.78	N/A	Baseline performance
SigExt (Fuzzy) + Llama-3	0.38	0.80	0.65	Original Paper's Method
SigExt (Semantic) + Llama-3	0.41	0.83	0.78	Our Extension

The projected data shows our semantic supervision method yields substantial gains not only in lexical overlap (ROUGE) but, more importantly, in semantic quality (BERTScore) and the model's ability to be steered (KIR).

Technical Deep Dive: Models and Training

The Extractor (SigExt Improved)

Model: markussagen/xlm-roberta-longformer-base-4096

Task: Fine-tuning for Token Classification / Sequence Labeling.

Key Consideration: Class Imbalance

- Salient sentences are rare. We will implement a **Weighted Cross-Entropy Loss** to heavily penalize errors on the positive class, ensuring the model learns effectively.

The Generator (LLM)

Model: Meta-Llama-3-8B-Instruct

Optimization: We will use **4-bit quantization (QLoRA)** via `bitsandbytes` to run this powerful model efficiently on available hardware like Google Colab GPUs.

Prompt Engineering

Example Italian Prompt Template

Sei un esperto di sintesi testuale. Leggi attentamente il seguente articolo. {source_text}

Inoltre, considera con particolare attenzione i seguenti concetti chiave estratti dal testo, che devono essere riflessi nel riassunto: {extracted_keyphrases}

Genera un riassunto astrattivo completo, coerente e fedele al testo originale.

The Path to Academic Excellence and Scientific Contribution

Strategic Strengths

- ✓ **Complexity:** Manages a sophisticated multi-model pipeline.
- ✓ **Novelty:** Introduces a superior semantic supervision method absent from the SOTA.
- ✓ **Relevance:** Addresses a course goal by bringing the research to an Italian dataset (WITS).
- ✓ **Publishability:** The “Fuzzy vs. Semantic” comparison in a new linguistic context is a compelling narrative for a scientific paper.

Next Steps

1. **Asset Acquisition:** Download WITS dataset, Llama-3, and XLM-R-Longformer checkpoints.
2. **Pipeline Implementation:** Develop the Python script for generating semantic labels using `sentence-transformers`.
3. **Environment Setup:** Configure the Hugging Face `Trainer` with Weighted Loss for extractor fine-tuning.
4. **Execution:** Launch initial experimental runs on the WITS dataset to validate the pipeline.